

Zen Nguyen

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TECHNICAL SKILLS

- **Backend & Systems:** FastAPI, Node.js, Express, RESTful APIs, gRPC, TCP Sockets
- **Infrastructure & DevOps:** Docker, AWS, Linux, Git
- **AI & Data:** RAG Architecture, Vector Databases, LangChain, OpenAI API, Gemini API, Pandas
- **Languages:** Python, Go, Java, C, C++, TypeScript, JavaScript, SQL, HTML/CSS

TECHNICAL PROJECTS

TitanSwarm (Autonomous AI Co-Pilot)

Jan 2026 - Present

Personal Project · Python, LangChain, RAG Architecture, Gemini API, Streamlit

- Automated end-to-end job application workflows by architecting an autonomous AI Co-Pilot using **Python** and **LangChain**, resulting in the discovery, extraction, and parsing of real-time Software Engineering postings.
- Delivered uniquely tailored, ATS-optimized PDF resumes by building a zero-hallucination **RAG Architecture** engine using **Gemini API** and a FAISS Vector Store, processing verified user data.
- Enabled users to review and manage applications across a full pipeline by designing and implementing a Streamlit Human-in-the-Loop dispatch UI with a Kanban board and persistent state via a SQLite repository.
- Validated all components including the scraper, AI tailor, PDF generator, and repository layer by applying Test-Driven Development (TDD) with a 113-test suite using Pytest, mocking, and async fixtures.

SFU Course Tracker (sfucourseplanner.me)

Nov 2025 - Jan 2026

Personal Project · Python, Docker, AWS, FastAPI, SQL

- Optimized deployments and reduced infrastructure costs by migrating a full-stack platform from Azure to **AWS** (EC2) and implementing **Docker** container orchestration.
- Increased data throughput by 10x while respecting server constraints by engineering a scraping pipeline using Asyncio and HTTPX with semaphore-based rate limiting.
- Designed and implemented a custom parser to tokenize and evaluate nested boolean prerequisite strings, converting unstructured text into a deterministic Abstract Syntax Tree (AST).
- Stored recursive tree structures to optimize read performance for complex queries by designing a normalized schema using SQLAlchemy with JSON-type columns.

TitanStore (Distributed Raft Database)

Jan 2026 - Present

Personal Project · Go, Raft Consensus Algorithm, TCP Sockets, PostgreSQL

- Developed a distributed key-value database in **Go** from scratch, implementing the **Raft consensus** protocol for leader election and fault-tolerant log replication across a cluster of nodes.
- Engineered a crash-safe Write-Ahead Log (WAL) with atomic snapshots to guarantee data durability and enable node recovery after failures without data loss.
- Designed and implemented a high-throughput raw **TCP** server on port 6001, parsing a custom text protocol supporting SET, GET, and ERR NOT_LEADER operations with non-blocking I/O.

EDUCATION

Bachelor of Science, Computing Science

May 2025 - Present

Simon Fraser University

- CGPA: 3.74 / 4.33
- Relevant Coursework: Discrete Math (A+), Computer Systems(A), Data Structures & Programming (B+), Linear Algebra(A-).
- Langara College | Associate of Science, Computer Science Jan 2024 - Apr 2025